Xuhao Luo

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Education

University of Illinois Urbana-Champaign Aug. 2021 - Nov. 2026 (Anticipated)
Ph.D. Candidate in Computer Science

University of California San Diego Sep. 2019 - Mar. 2021
M.S. in Computer Science

University of Science and Technology of China Sep. 2015 - Jun. 2019
B.S. in Applied Physics

Research Publication

- **Xuhao Luo**, Emaan Atique, Aishwarya Ganesan, Ramnatthan Alagappan, **SkyLog: Reducing Cost and Latency in a Multi-Cloud Shared Log** (In submission to NSDI)
- **Xuhao Luo**, Shreesha G. Bhat*, Jiyu Hu*(*equal contribution), Aishwarya Ganesan, Ramnatthan Alagappan, **LazyLog: A New Shared Log Abstraction for Low-Latency Applications** (***SOSP 2024***) ***Best Paper Award, Invited to ACM ToCS***
- **Xuhao Luo**, Ramnatthan Alagappan, Aishwarya Ganesan, **SplitFT: Fault Tolerance for Disaggregated Datacenters via Remote Memory Logging** (***EuroSys 2024***)
- **Xuhao Luo**, Weihai Shen, Shuai Mu, Tianyin Xu, **DepFast: Orchestrating Code of Quorum Systems** (***USENIX ATC 2022***)
- Shreesha G. Bhat, Tony Hong, **Xuhao Luo**, Jiyu Hu, Aishwarya Ganesan, Ramnatthan Alagappan, **Low End-to-End Latency atop a Speculative Shared Log with Fix-Ante Ordering** (***OSDI 2025***)
- Zhiyuan Guo*, Yizhou Shan*(*co-first author), **Xuhao Luo**, Yutong Huang, Yiying Zhang, **Clio: A Hardware-Software Co-Designed Disaggregated Memory System** (***ASPLOS 2022***)

Experience

University of Illinois Urbana-Champaign May. 2021 - Now
Research Assistant *Urbana, IL, USA*

- *LazyLog*: Built an RDMA-based shared log with μs-level append latency. Decoupled appends from ordering across shards to support latency-critical distributed applications; received the SOSP 2024 Best Paper Award.
- *SplitFT*: Built fault-tolerance support for disaggregated datacenters using RDMA to replicate write-ahead logs in remote memory. Achieved near-zero performance loss relative to a local file system for the evaluated applications.
- *SkyLog*: Built a multi-cloud shared log supporting seamless reconfiguration and reducing cross-cloud data-transfer costs.
- *DepFast*: Built a C++ coroutine-based framework to structure distributed protocol execution and simplify implementation and reasoning about fail-slow tolerance.

Google May. 2026 - Aug. 2026
Software Engineer Intern, Host: Xiaozhou (Steve) Li *Sunnyvale, CA, USA*

- Designed and implemented a production-deployed framework for user-defined functions in Bigtable compaction, demonstrating that in-system execution could replace selected data-processing pipelines.

Amazon Web Service May. 2024 - Aug. 2024
Applied Scientist Intern, Mentor: Prof. George Amvrosiadis, Visiting Scholar *Seattle, WA, USA*

- Investigated and improved space reclamation in AWS S3 ShardStore; built a tool to evaluate reclamation policies.

Amazon Web Service May. 2022 - Aug. 2022
Applied Scientist Intern, Mentor: Shen Li, Principal Engineer *Seattle, WA, USA*

- Improved the performance and reliability of the volume metadata updating workflow for AWS S3 volume metadata cache service with Amazon Quantum Ledger Database (QLDB).

Microsoft Research Jun. 2020 - Sep. 2020
Research Intern *Beijing, China*

- Designed and implemented runtime support for distributed ML: task scheduling and dispatch, plus a CUDA-based inter-GPU communication layer for multi-GPU execution.
- Implemented ring AllGather over NVLink in a multi-process GPU setup, using host memory for synchronization.

University of California San Diego Sep. 2019 - Dec. 2020
Research Assistant, advised by Prof. Yiying Zhang *La Jolla, CA, USA*

· *Clio*: Contributed to an FPGA-based, hardware/software co-designed disaggregated-memory system, including network-stack design and implementation for remote-memory access.

University of Science and Technology of China Jan. 2019 - May. 2019
Graduation Project, Advisor: Prof. Xi Jin

· Integrated a custom FPGA accelerator with TensorFlow through a C++ interface using OpenCL, connecting ML framework execution to accelerator hardware.

· Designed and implemented an 8-bit quantized convolution module on a Xilinx Virtex FPGA. On ResNet-50, achieved 5.17x speedup and 66% lower memory usage versus CPU TensorFlow on Xeon E5 2686.

Skills

Language	C/C++, Python, Go, Java, Rust, Haskell, Verilog
Tools/Framework	RDMA, TensorFlow, vLLM, SGLang, CUDA, Docker, Zookeeper

Honors and Awards

- SOSP'24 Best Paper Award Nov 2024
- SOSP'24 Student Travel Grant Nov 2024
- EuroSys'24 Student Travel Grant Apr 2024
- ASPLOS'22 Student Travel Grant Feb 2022
- USTC Class of 2019 Outstanding Graduates May 2019

Services

- EuroSys'26 Shadow PC
- OSDI'23 Artifact Evaluation Committee
- USENIX ATC'23 Artifact Evaluation Committee
- ACM TAAS (Transactions on Autonomous and Adaptive Systems) Invited Reviewer